Electronic Principles and Devices · Unit_01 · Unit-1 Semiconductor Diode

DRAFT — UE26EC141A · Unit_01 · Unit-1 Semiconductor Diode · Diode equivalent circuits Series diode configurations

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Diode equivalent circuits Series diode configurations

1 Diode equivalent circuits Series diode configurations

Study focus: Diode equivalent circuits Series diode configurations. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000046 · EV0000047

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today's topic will be diode equivalent circuit or diode approximation and series diode configuration.

Key details

- So coming to the diode approximation, is a mathematical method used to approximate the nonlinear behavior of the real diode to enable the calculations and circuit analysis.
- There are three different approximations used to analyze the diode circuits, namely the Piecewise linear equivalent circuits,.
- Simplified diode characteristics, and Ideal diode characteristics.
- Piecewise linear diode characteristics.
- Under forward bias, with respect to the V-I characteristics, the diode behaves as a practical voltage source with a magnitude equal to the knee voltage in series with the diode average series with the diode average AC resistance.
- This is shown the representation.
- Under reverse bias, the diode is an open switch, and no current flows here through it, act like an ideal diode reverse characteristics.
- The voltage drop across the diode is calculated by the formula VD is equal to Vb plus ID into RB.
- Coming to the Simplified diode characteristics, under forward bias, the diode turns on when a voltage VK or VB is applied across it.
- It turns off if the voltage applied is less than the VK.
- Thus, diode is represented by voltage of VK in a series with a closed path switch, as shown.
- Therefore, VD is equal to VB.

- Then coming to the reverse bias, it is an open switch similar to that of a practical reverse bias voltage.
- Coming to the Ideal diode characteristics here, we already know the diode is considered as a closed switch with a zero voltage drop.
- Then when in forward bias VD and open switch when it's in reverse bias.
- When the diode is connected in series, we call it as a diode configuration, a series diode configuration with a resistor.
- It is assumed that the forward characteristics or forward resistance of the diode is usually so small compared to the other series elements of the network that can be ignored.
- In general, the diode is in on state if the current established by the applied sources is such that its direction matches that of an arrow in the diode symbol and applied voltage across the diode is silicon VD greater than 0.7, germanium VD greater than 0.3, and gallium arsenide VD should be greater than 1.2 volts.
- And when it's reverse bias, diode is reverse connected to the positive voltage here and represented as an open circuit, and hence ID is current is zero here, and VR also is equal to zero volts.
- This is with respect to series diode configuration with a resistor.
- Assume the LED voltage as 0.7.
- We have a silicon and a LED here, and we have an output resistance 680 ohm.
- Then we need to apply the KVL here.
- We need to remember the KVL equation, which we learned from 12th standard.
- The sum of all the voltages across the closed loop network is always equal to zero.

Example or application from the lecture

- Let us see one by one today.
- Let us solve one numerical based on the series diode configuration.
- The question is, determine the Vout and ID for a given series diode circuit.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 05:00–09:59

V is equal to E minus VK1 minus VK2.

Key details

- We get 9.5 volts output.
- Then how to calculate ID?
- Since we know the voltage across this, so VR by R.
- That is Vout by R.
- Vout 9.5 divided by 680 ohm, which is 13.97 milliamps.

Example or application from the lecture

- So any numerical given in this fashion, if the diode is connected in series, we can calculate by applying a KVL first, calculate the output voltage, and take that output voltage divided by

the total resistance, we get a current flowing through it.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Diode equivalent circuits Series diode configurations transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/006_Diode_equivalent_circuits_Series_diode_configurations__91f64c479cb44c60.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode equivalent circuits Series diode configurations transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/006_Diode_equivalent_circuits_Series_diode_configurations__91f64c479cb44c60.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode equivalent circuits Series diode configurations transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/006_Diode_equivalent_circuits_Series_diode_configurations__91f64c479cb44c60.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9966; only the JSON segmentation supplies the displayed timestamps.
Diode equivalent circuits Series diode configurations transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/006_Diode_equivalent_circuits_Series_diode_configurations__91f64c479cb44c60.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00022	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0064. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0065 to s0075. Sidecar variants compare...
SRC00023	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0064. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0065 to s0075. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0064. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0065 to s0075. Sidecar variants compare...
SRC00025	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0064. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0065 to s0075. Sidecar variants compare...

C Revision and retrieval practice